```
#include <WiFi.h>
#include <WebServer.h>
#include <Wire.h>
#include <Adafruit_PWMServoDriver.h>

// --- Network Setup ---
const char* ssid = "AgriRobot_AP";
const char* password = "12345678";

IPAddress local_IP(192, 168, 4, 1);
IPAddress gateway(192, 168, 4, 1);
IPAddress subnet(255, 255, 255, 0);

WebServer server(80);

// --- Component Setup ---
Adafruit_PWMServoDriver pwm = Adafruit_PWMServoDriver(0x40); // 0x40 is default I2C
address

#define SERVOMIN  150 // PCA9685 minimum pulse length
#define SERVOMAX  600 // PCA9685 maximum pulse length

// L298N Motor Driver Pins
const int IN1 = 14;
const int IN2 = 27;
const int IN3 = 26;
const int IN4 = 25;

// Note: Relay 1 is wired to PCA9685 Channel 14
// Note: Relay 2 is wired to PCA9685 Channel 15

// --- Web Interface HTML & JS ---
const char INDEX_HTML[] PROGMEM = R"rawliteral(
<!DOCTYPE HTML><html>
<head>
  <meta name="viewport" content="width=device-width, initial-scale=1, maximum-scale=1,
user-scalable=no">
  <title>Rover Dashboard</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; background-color: #121212; color:
white; margin: 0; padding: 10px; }
    h2 { color: #00ffcc; }
    .card { background-color: #1e1e1e; padding: 15px; border-radius: 10px; margin-bottom: 20px;
}
```

```
    img { width: 100%; max-width: 320px; border: 2px solid #00ffcc; border-radius: 5px; }

    /* CSS Fixes for Hold-to-Move Buttons */
    .btn {
      background-color: #333; color: white; border: none; padding: 12px 20px;
      font-size: 16px; margin: 5px; border-radius: 5px; cursor: pointer;
      user-select: none;
      -webkit-user-select: none;
      touch-action: manipulation;
      -webkit-touch-callout: none;
    }
    .btn:active { background-color: #00ffcc; color: black; }

    .dpad { display: grid; grid-template-columns: repeat(3, 1fr); gap: 5px; max-width: 250px;
margin: 0 auto; }
    .empty { visibility: hidden; }
    input[type=range] { width: 90%; max-width: 300px; margin: 10px 0; }
  </style>
</head>
<body>
  <h2>AgriRobot Command Center</h2>

  <!-- Camera Section -->
  <div class="card">
    <h3>Live Feed</h3>
    <img id="cam" src="" alt="Connecting to Camera...">
    <br>
    <button class="btn" onclick="fetch('http://192.168.4.50:82/flashon')">Flash ON</button>
    <button class="btn" onclick="fetch('http://192.168.4.50:82/flashoff')">Flash OFF</button>
  </div>

  <!-- Movement Section -->
  <div class="card">
    <h3>Drive Control</h3>
    <div class="dpad">
      <div class="empty"></div>
      <button class="btn"
        onmousedown="move('F')" onmouseup="move('S')" onmouseleave="move('S')"
        ontouchstart="event.preventDefault(); move('F')" ontouchend="event.preventDefault();
move('S')">Forward</button>
      <div class="empty"></div>

      <button class="btn"
        onmousedown="move('L')" onmouseup="move('S')" onmouseleave="move('S')"
```

```
        ontouchstart="event.preventDefault(); move('L')" ontouchend="event.preventDefault();
move('S')">Left</button>

      <button class="btn" onclick="move('S')" ontouchstart="event.preventDefault();
move('S')">Stop</button>

      <button class="btn"
        onmousedown="move('R')" onmouseup="move('S')" onmouseleave="move('S')"
        ontouchstart="event.preventDefault(); move('R')" ontouchend="event.preventDefault();
move('S')">Right</button>

      <div class="empty"></div>
      <button class="btn"
        onmousedown="move('B')" onmouseup="move('S')" onmouseleave="move('S')"
        ontouchstart="event.preventDefault(); move('B')" ontouchend="event.preventDefault();
move('S')">Back</button>
      <div class="empty"></div>
    </div>
  </div>

  <!-- Arm Section -->
  <div class="card">
    <h3>Robotic Arm (PCA9685)</h3>
    Servo 0 (Base): <input type="range" min="0" max="180" value="90" oninput="setServo(0,
this.value)"><br>
    Servo 1 (Shoulder): <input type="range" min="0" max="180" value="90" oninput="setServo(1,
this.value)"><br>
    Servo 2 (Elbow): <input type="range" min="0" max="180" value="90" oninput="setServo(2,
this.value)"><br>
    Servo 3 (Wrist Pitch): <input type="range" min="0" max="180" value="90"
oninput="setServo(3, this.value)"><br>
    Servo 4 (Wrist Roll): <input type="range" min="0" max="180" value="90"
oninput="setServo(4, this.value)"><br>
    Servo 5 (Gripper): <input type="range" min="0" max="180" value="90" oninput="setServo(5,
this.value)">
  </div>

  <!-- Relay Section -->
  <div class="card">
    <h3>Accessories</h3>
    <button class="btn" onclick="toggleRelay(1, 1)">Relay 1 ON</button>
    <button class="btn" onclick="toggleRelay(1, 0)">Relay 1 OFF</button><br>
    <button class="btn" onclick="toggleRelay(2, 1)">Relay 2 ON</button>
    <button class="btn" onclick="toggleRelay(2, 0)">Relay 2 OFF</button>
```

```
  </div>

  <script>
    // --- HIGH-SPEED FRAME ENGINE ---
    const camEl = document.getElementById('cam');
    const streamUrl = 'http://192.168.4.50/capture?t=';

    function fetchNextFrame() {
      const img = new Image();
      img.onload = () => {
        requestAnimationFrame(() => {
          camEl.src = img.src;
          fetchNextFrame();
        });
      };
      img.onerror = () => {
        setTimeout(fetchNextFrame, 50);
      };
      img.src = streamUrl + Date.now();
    }

    fetchNextFrame();

    // --- ROVER CONTROLS ---
    function move(dir) { fetch(`/move?dir=${dir}`); }
    function setServo(id, val) { fetch(`/servo?id=${id}&val=${val}`); }
    function toggleRelay(id, state) { fetch(`/relay?id=${id}&state=${state}`); }
  </script>
</body>
</html>
)rawliteral";

// --- API Handlers ---

void handleRoot() {
  server.send(200, "text/html", INDEX_HTML);
}

void handleMove() {
  if (server.hasArg("dir")) {
    String dir = server.arg("dir");
    if (dir == "F") {
      digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
      digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
```

```
    }
    else if (dir == "B") {
      digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
      digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
    }
    else if (dir == "L") {
      digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
      digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
    }
    else if (dir == "R") {
      digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
      digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
    }
    else if (dir == "S") {
      digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);
      digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);
    }
    server.send(200, "text/plain", "OK");
  }
}

void handleServo() {
  if (server.hasArg("id") && server.hasArg("val")) {
    int id = server.arg("id").toInt();
    int angle = server.arg("val").toInt();

    int pulse = map(angle, 0, 180, SERVOMIN, SERVOMAX);
    pwm.setPWM(id, 0, pulse);

    server.send(200, "text/plain", "OK");
  }
}

void handleRelay() {
  if (server.hasArg("id") && server.hasArg("state")) {
    int id = server.arg("id").toInt();
    int state = server.arg("state").toInt();

    // Assign Relay 1 to PCA channel 14, Relay 2 to PCA channel 15
    int channel = (id == 1) ? 14 : 15;

    // Standard Relay Logic via PCA9685
    // Assuming relays are Active-LOW (pulling pin to GND/0 turns them ON).
    if (state == 1) {
```

```
    pwm.setPWM(channel, 0, 4096); // Pin LOW -> Relay ON
  } else {
    pwm.setPWM(channel, 4096, 0); // Pin HIGH -> Relay OFF
  }

  server.send(200, "text/plain", "OK");
 }
}

// --- Setup & Loop ---

void setup() {
 Serial.begin(115200);
 delay(1000); // Give serial monitor time to open

 // Init L298N Motor Pins
 pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT);
 pinMode(IN3, OUTPUT); pinMode(IN4, OUTPUT);
 digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);
 digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);

 // --- I2C DIAGNOSTIC TEST ---
 Wire.begin(21, 22);
 Serial.println("\n--- HARDWARE DIAGNOSTICS ---");
 Serial.println("Checking for PCA9685 Servo Driver...");

 Wire.beginTransmission(0x40);
 byte error = Wire.endTransmission();

 if (error == 0) {
  Serial.println("SUCCESS: PCA9685 found at address 0x40!");
 } else {
  Serial.println("ERROR: PCA9685 NOT FOUND.");
  Serial.println("-> Check: Are SDA(21) and SCL(22) swapped?");
  Serial.println("-> Check: Is VCC powered?");
 }
 Serial.println("----------------------------\n");

 // Init PCA9685
 pwm.begin();
 pwm.setPWMFreq(50); // Analog servos run at ~50 Hz

 // Center all 6 servos to 90 degrees initially
 for (int i = 0; i < 6; i++) {
```

```
    pwm.setPWM(i, 0, map(90, 0, 180, SERVOMIN, SERVOMAX));
  }

  // Force Relays (Channels 14 & 15) to turn OFF on boot
  // Pulling pins HIGH to turn off Active-LOW relays.
  pwm.setPWM(14, 4096, 0);
  pwm.setPWM(15, 4096, 0);

  // Init Access Point
  WiFi.softAPConfig(local_IP, gateway, subnet);
  WiFi.softAP(ssid, password);
  Serial.print("Main Rover AP Started. IP Address: ");
  Serial.println(WiFi.softAPIP());

  // Init Webserver Routes
  server.on("/", handleRoot);
  server.on("/move", handleMove);
  server.on("/servo", handleServo);
  server.on("/relay", handleRelay);

  // Handle CORS for browser
  server.enableCORS(true);
  server.begin();
}

void loop() {
  server.handleClient();
}
```